Code No: 123CT **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, March - 2022
DIGITAL LOGIC DESIGN
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Write about the different number system representations and Evaluate the 2's and 1's complement of the binary number 1010110.
b) Given the Boolean function $F = x\,y + x'y' + y'z$. Implement it with AND, OR, and NOT gates. [6+9]

2.a) Convert the number $(0.875)_{10}$ into its i) Binary ii) trinary iii) quinary.
b) Determine the canonical sum-of-products representation of the below functions.
f(x,y,z) = z + $(x' + y)\ (x + y')$ [9+6]

3. Discuss about the three variable, four variable K-maps and simplify the function using Karnaugh map method F (A,B,C,D) = Σ (4,5,7,12,14,15)+ Σd(3,8,10). [15]

4.a) Implement the function F(A,B,C,D) = ∑(0,1,2,3,4,8,9,12) with NAND-AND and also AND-NOR logic implementations
b) Design decimal adder and draw its logic diagram. [8+7]

5.a) Mention about the different adders and subtractors used in digital design and explain the working and design of 4 - bit carry look-ahead adder circuit.
b) Design magnitude comparator and explain its working. [9+6]

6.a) Design a 4 bit parallel-in serial-out shift register and explain its operation.
b) Draw the schematic of Toggle-flip-flop (T) Give its truth-table and justify these entries. [9+6]

7.a) Write about the 4-bit Johnson counter using D-flip flop with its operation.
b) Write about the state reduction and role free conditions in asynchronous sequential circuits. [8+7]

8.a) Differentiate between the sequential memory and cache memory.
b) Write about the Programmable Logic Arrays and their use in digital design.
c) What are the different types of ROMs and compare them. [5+5+5]

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